SceneFormer: Neural Architecture Search of Transformers for Remote Sensing Scene Classification

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***Abstract*— Deep learning-based scene classification methods have long been a key research area in remote sensing imagery due to their wide-ranging applications. Recently, Transformer models have achieved significant progress in computer vision, making vision transformers (ViTs) a promising direction for scene classification. However, the spatial complexity of remote sensing imagery poses unique challenges for applying Transformers directly. Manually designing Transformers tailored for remote sensing scene classification is time-consuming under model parameter constraints and requires extensive domain expertise. To address this challenge, neural architecture search (NAS) methods provide an effective solution to construct optimal Transformer architectures for remote sensing scene classification automatically. In this work, we propose SceneFormer, an automated Transformer architecture search framework tailored for scene classification tasks. In SceneFormer, we construct a dedicated search space to search for the optimal Transformer. Moreover, we design a supernet training strategy to train numerous candidate architectures within the search space simultaneously. Furthermore, SceneFormer employs the evolutionary search to find the optimal Transformer architecture under specific resource constraints. Experiments on three high-spatial-resolution (HSR) datasets demonstrate the effectiveness of SceneFormer.**

***Index Terms*— Evolutionary algorithm, neural architecture search (NAS), remote sensing, scene classification, vision transformer (ViT).**

NOMENCLATURE

CNN	Convolutional neural network.
Flops	Floating-point operations.
HSR	High spatial resolution.
Kappa	Cohen's Kappa coefficient, a key statistical measure used to assess classification accuracy or evaluator consistency.
MHSA	Multihead self-attention.
MLP	Multilayer perceptron.
NAS	Neural architecture search.
OA	Overall accuracy.
ViTs	Vision transformers.

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I. INTRODUCTION

REMOTE sensing scene classification plays a crucial role in various applications, including urban planning, land-use mapping, and image retrieval [1], [2]. Compared to pixel- or object-level classification, scene classification provides a more abstract and comprehensive interpretation of aerial images [3]. Unlike pixel-level classification and object detection, it emphasizes semantic labels, capturing higher level social and contextual attributes [4]. With the rise of deep learning, models like CNNs have demonstrated remarkable capabilities in extracting high-dimensional semantic information from HSR images [5], [6], [7]. Recently, ViTs have emerged as a promising alternative in computer vision, achieving state-of-the-art performance in natural image classification [8], [9], [10], [11]. Due to their ability to capture global dependencies, ViTs have also gained increasing attention in remote sensing scene classification. Several studies have explored ViT-based architectures for remote sensing tasks, highlighting their advantages over CNNs [12], [13], [14], [15], [16], [17]. These findings underscore the potential of Transformers for remote sensing scene classification.

Although ViTs have achieved notable success in natural image tasks, they encounter unique challenges when applied to HSR remote sensing imagery. The spatial complexity and diversity of HSR remote sensing images require models with greater representational capabilities. Designing Transformer architectures for remote sensing scene classification manually requires significant expertise and extensive experimentation. NAS, an automated method for designing neural network architectures, offers a powerful solution to these challenges [18]. By eliminating the dependence on deep domain knowledge, NAS enables the efficient design of Transformer architectures optimized for remote sensing tasks.

In this work, we propose SceneFormer, a ViT architecture search framework specifically designed for remote sensing scene classification tasks. Our research tackles two primary challenges.

1) How can ViT architectures be automatically designed to suit remote sensing scene classification?

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2) How can the automated search for ViT architectures be performed efficiently and effectively under model parameter constraints?

To address remote sensing scene classification tasks, we have constructed a search space for the ViT architecture to automatically search for the optimal Transformer architecture. We aim to balance and integrate the structural hyperparameters of ViT architectures. Inspired by AutoFormer [19], we set the search target as heterogeneous ViT architectures across layers. This involves searching for the overall network depth and embedding dimensions, as well as the MLP ratios and the number of attention heads for each layer. Compared to traditional Transformers [9], [10] that stack identical layers, SceneFormer designs different attention mechanisms for each layer. This approach better captures the multiscale features of remote sensing images and improves adaptability to complex scenes. Moreover, with optimization through NAS methods, heterogeneous architectures effectively allocate computational resources across layers and minimize redundant information processing. Ultimately, SceneFormer can automatically design Transformer architectures that better satisfy the needs of remote sensing scene classification tasks.

To improve efficiency, our supernet training strategy is designed to simultaneously train a large number of candidate architectures within the search space. When candidate architectures are evaluated, they can directly inherit trained weights from the supernet, eliminating the need for retraining and significantly enhancing the efficiency of the search process. First, we pretrain our supernet on natural image datasets to fully leverage the powerful representation capability of Transformer architectures and learn general image features. Then, we finetune the supernet on HSR datasets to adapt the features for remote sensing scene classification. This approach helps alleviate overfitting issues from small datasets. By combining the general features learned from natural images with the specific demands of remote sensing data, our pretraining and finetuning strategy enhances both the performance and efficiency of the classification task.

To further consider the constraints of the model parameter, we employ an evolutionary search to explore promising Transformer architectures within the search space. Evolutionary algorithms are known for their strong global optimization capability in complex, high-dimensional spaces. They can effectively avoid local optima and better support flexible architecture designs, including irregular stacking patterns of Transformer layers. Moreover, evolutionary algorithms generate diverse candidate architectures through crossover and mutation operations. Eventually, the best candidate architecture is selected based on its ability to meet specific remote sensing scene classification requirements while balancing performance and resource constraints.

In summary, the main contributions of this article are summarized below.

1) We propose an automated Transformer architecture search method for remote sensing scene classification tasks.
2) By combining pretraining and finetuning, our supernet training strategy ensures that the model learns general image features efficiently while adapting to remote sensing tasks.
3) We employ an evolutionary algorithm to effectively explore the entire search space and find the optimal Transformer architecture under specific resource constraints.

II. Related Work

Recent advancements in ViTs have significantly enhanced remote sensing image classification by capturing long-range dependencies and global contextual information. However, designing optimal Transformer architectures for remote sensing tasks remains challenging due to the need to balance classification accuracy, efficiency, and model complexity. To address this, NAS has emerged as an automated solution that efficiently explores model architectures and improves performance. In this section, we review the key developments of ViTs in the field of computer vision, and the progression of NAS techniques, and provide a brief introduction to the evolutionary algorithms.

A. Vision Transformers

The Transformer model was originally designed for natural language processing tasks [8], but in recent years, the ViT has made significant advancements in the field of computer vision [9], [10], [11]. These breakthroughs demonstrate the effectiveness of Transformer architectures in processing visual data. In computer vision tasks, especially in image classification, object detection, and semantic segmentation, researchers have developed a variety of Transformer-based models.

The ViT first splits the input image into several patches, then flattens each patch into a vector and maps it to a patch token via a linear transformation. A positional encoding is added to each patch token to retain the spatial position information. These patch tokens are then input into the Transformer encoder. The encoder models the relationships between patches using the MHSA mechanism. MHSA allows each patch to interact with others and capture both global and local features of the image. The Transformer encoder also further processes the feature vectors through an MLP to extract higher level features. Finally, the output of the stacked Transformer encoders is used for downstream tasks, such as classification or object detection.

In recent years, there has been a growing interest in applying ViTs to remote sensing image classification, particularly for scene classification tasks. The ability of ViTs to capture long-range dependencies and global contextual information makes them well-suited for handling complex spatial patterns in remote sensing images. This has led to several works exploring the potential of ViTs for remote sensing scene classification, with notable advancements in improving model efficiency and robustness for high-resolution and multitemporal data. Bazi et al. [13] apply the ViT model to remote sensing scene classification and leverage data augmentation

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to enhance accuracy. TRS [14] adopts a progressive CNN-to-Transformer design and introduces the MHSA-Bottleneck, which replaces 3×3 convolutions with MHSA for improved efficiency. SCViT [15] considers both geometric and channel information for scene classification in complex remote sensing scenarios. CTNet [16] combines CNNs and Transformers using a joint loss function, further enhancing feature representation capability. HFFT-PD [17] proposes a general multilevel feature enhancement method that can be easily integrated into various hierarchical transformer models, achieving excellent performance in remote sensing scene classification tasks. These efforts highlight the promise of ViTs in addressing the unique challenges posed by remote sensing imagery.

B. Neural Architecture Search

NAS is an automated deep learning model design approach [18]. It defines a search space and employs an appropriate search method to find the optimal neural network architecture within that space. The goal of NAS methods is to automatically search for an architecture that achieves the best performance on the validation set after sufficient training [20]. This goal can be formally expressed as

$$w_a = \arg\min_{w} \mathcal{L}_{\text{train}}(\mathcal{N}(a, w)) \tag{1}$$

$$a^* = \arg\max_{a \in \mathcal{A}} \text{ACC}_{\text{val}}(\mathcal{N}(a, w_a)) \tag{2}$$

where $\mathcal{A}$ represents the search space, and $\mathcal{N}(a, w)$ denotes a neural architecture a with its associated training weights w. $\mathcal{L}_{\text{train}}(\cdot)$ refers to the loss function on the training set, and $\text{ACC}_{\text{val}}(\cdot)$ indicates the accuracy on the validation set.

Although NAS can automatically design neural network architectures, it typically requires significant computational resources. Therefore, researchers continuously propose new optimization techniques to enhance search efficiency and reduce computational costs [21], [22], [23], [24]. Recent studies have shown that one-shot model-based NAS methods [19], [20], [25], [26], [27] can accelerate the search process and reduce computational overhead. The one-shot model trains a supernet, which contains multiple subnets (or candidate architectures)

$$W_{\mathcal{A}} = \arg\min_{W} \mathcal{L}_{\text{train}}(\mathcal{N}(\mathcal{A}, W)). \tag{3}$$

When evaluating the performance of a candidate architecture, it directly inherits the corresponding weights from the supernet and is tested on the validation set

$$a^* = \arg\max_{a \in \mathcal{A}} \text{acC}_{\text{val}}(\mathcal{N}(a, W_{\mathcal{A}}(a))) \tag{4}$$

where $W_{\mathcal{A}}(a)$ represents the corresponding parts of the weights from the supernet.

This weight-sharing mechanism significantly reduces computational costs. Various search algorithms can be applied within this framework. Evolutionary algorithms, reinforcement learning, or gradient-based methods are widely used during the search process.

Previous NAS methods have primarily focused on CNN architectures. However, with the rise of Transformers, recent NAS approaches have shifted toward exploring Transformer architectures or integrating them with CNNs. Glit [24] incorporates a locality module to capture local correlations in images and proposes a hierarchical search strategy to improve the search results. AutoFormer [19] introduces a weight entanglement strategy for training a supernet that covers a wide range of subnets and employs an evolutionary algorithm to identify promising Transformer architectures. TF-TAS [28] and AZ-NAS [29] leverage zero-cost proxies for training-free Transformer architecture search.

Recent advancements in NAS have also been applied to remote sensing scene classification, demonstrating the potential of automatic architecture design for this domain. RSNet [30] introduces a two-stage cascade optimization strategy and designs a switchable recognition module to search for deep neural networks tailored for various HRS imagery recognition tasks. SceneNet [31] proposes an EA-based NAS method for remote sensing image scene classification, achieving an adaptive balance between computational complexity and the performance tradeoffs of the searched architectures. E2SCNet [32] constructs a diverse search space using eight lightweight operators and employs multiobjective evolutionary search with a supernet-based weight-sharing strategy to achieve efficient architecture search.

C. Evolutionary Algorithm

Evolutionary algorithms [33] are a class of optimization methods inspired by natural selection. They are widely employed to tackle complex optimization problems. The core idea of evolutionary algorithms is to iteratively improve solutions by simulating the process of biological evolution. In these algorithms, a population of potential solutions generates new solutions through crossover and mutation operations. The best-performing individuals are then selected based on their fitness evaluation and used to reproduce the next generation. This process continues until a satisfactory solution is found or a predefined stopping condition is met.

Evolutionary algorithms can efficiently address multiobjective optimization problems and perform well under resource constraints. In NAS methods, evolutionary algorithms optimize model performance while minimizing resource consumption, which is beneficial for automatically designing efficient and lightweight neural network architectures. In recent years, many studies [19], [20], [27], [34] have combined evolutionary algorithms with NAS to design efficient and generalizable neural network architectures.

III. SceneFormer

For remote sensing scene classification, SceneFormer is proposed to automatically search for a ViT architecture. Initially, the search space of the Transformer architecture is defined. In the supernet training stage, the supernet training strategy is introduced to train numerous candidate architectures within the search space simultaneously. In the evolutionary search stage, we propose an evolutionary search to obtain the optimal Transformer architecture that can satisfy the model parameter constraint. The overall process is shown in the

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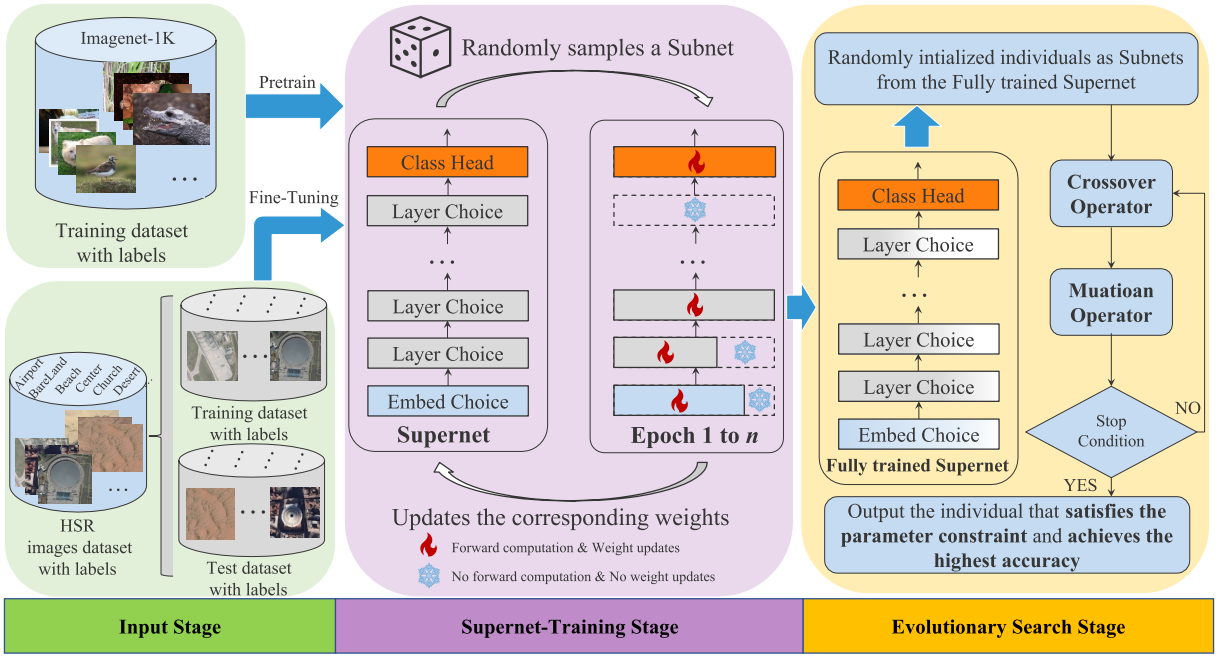


Fig. 1. Overall process of SceneFormer.

TABLE I
DETAILED CONFIGURATION OF THE SEARCH SPACE

Embed dimension	Depth	Head numbers	MLP ratio
(192, 216, 240)	(12, 13, 14)	(3, 4)	(3.5, 4.0)

Fig. 1. The details of the search process are explained in Sections III-A–III-C.

A. Search Space

We search not only for the overall depth of the network and the embedding layer dimensions but also for the specific structure of each layer. Typically, manually designed ViT architectures stack identical layer structures. In this work, we search for a ViT architecture with heterogeneous structures across layers. Based on the ViT architecture [9], we search the MLP ratio and the number of attention heads for each layer. The detailed configuration of the search space is provided in Table I.

Each layer equips multiple attention heads to learn different features in parallel, thus enhancing the model's ability to capture information. Uniform MHSA blocks cannot adjust the number of heads based on the distinct feature needs of different layers. This limitation may affect the expressive power of some layers. We search for the optimal number of attention heads to better capture the features of each layer. Additionally, we improve overall performance and efficiency by reducing redundant attention heads in some layers.

Each layer's MLP block consists of two fully connected layers. The first layer expands the input dimensions to a higher hidden dimension. This enhances the model's expressive power. The second layer restores the output dimensions to the input dimensions. In the ViT, the hidden dimensions of each layer are typically set to four times the input dimensions.

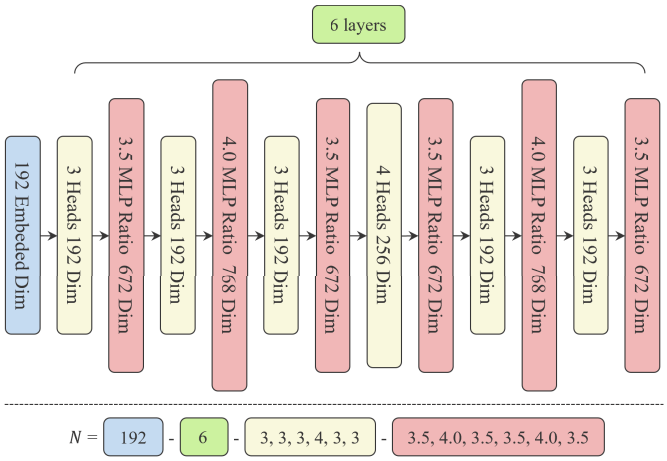


Fig. 2. Example of individual encoding.

However, different layers have different needs for feature extraction. Therefore, using the same MLP block may not be the best choice. To enable residual connections, the input and output dimensions of each layer usually need to remain consistent. Therefore, we search for the hidden dimensions of the MLP blocks (i.e., the MLP ratio) for each layer to create unique MLP blocks.

We map the Transformer architecture to an individual encoding

$$(d, m_1, m_2, \ldots, m_d, h_1, h_2, \ldots, h_d, e). \tag{5}$$

The first element d represents the depth of the network. The following d elements $(m_1, m_2, \ldots, m_d)$ represent the MLP ratio for each of the d layers. Similarly, $(h_1, h_2, \ldots, h_d)$ represent the number of attention heads in each layer, and the last element e denotes the embedding dimension of the

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network. With this encoding, each architecture in the search space corresponds one-to-one with an individual encoding. Fig. 2 shows an example of individual encoding.

B. Supernet

During the search process, a large number of candidate architectures in the search space need to be evaluated. Traditional NAS methods train each candidate architecture independently. However, this incurs extremely high-computational costs in large-scale search spaces. To address this issue, we follow Autoformer [19] and train a supernet that covers the entire search space to simultaneously train all subnets within the search space. This eliminates the need to retrain a large number of candidate architectures. All structural hyperparameters of the supernet are set to the maximum values defined in the search space. All candidate architectures can be represented as subnets of the supernet. The specific structure of the supernet is shown in Fig. 3.

In the subsequent search process, we allow subnets to inherit the corresponding weights from the supernet and evaluate them on the validation set. It is worth noting that, as shown in the solid section of Fig. 3, the subnet always inherits the initial part of the supernet along with its training weights. Specifically, for the FC layers, the subnet prioritizes inheriting the front portion of the FC layers rather than inheriting the random part. In particular, for the qkv vectors involved in the attention calculation, we split them into parts for each attention head, with the subnet prioritizing the first several attention heads. Then, for each attention head, the subnet concentrates on inheriting the front part. For the number of network layers, the subnet prioritizes inheriting the initial few Transformer layers. The supernet training process is shown in Fig. 4.

We first train the supernet on the general natural image dataset ImageNet-1k [35], and then we finetune the supernet on the HSR images dataset [3], [36], [37]. We first preprocess the image resolution to 224 × 224 to align with the standard input size of ViTs. This preprocessing step effectively reduces computational costs while maintaining high accuracy. The Transformer performs well in processing global information and effectively extracts semantic features without being affected by local details.

During the training of the supernet, we sample a subnet from the supernet at each epoch and update its corresponding weights. This process can be formally described as

$$W_{\mathcal{A}} = \arg\min_{W} E_{a\sim U(\mathcal{A})}[\mathcal{L}_{\text{train}}(\mathcal{N}(a, W(a)))] \tag{6}$$

where U denotes the uniform distribution. Equation (6) is a specific implementation of (3). Algorithm 1 shows the specific process of supernet training.

Under this training strategy, the weights of the front part of each structure are updated most frequently. Since the subnet always prioritizes inheriting the front parts of the supernet's structures, the subnet that inherits the supernet's weights can ultimately achieve performance comparable to that of training from scratch.

Algorithm 1 Supernet Training Procedure

Input: Search space $\mathcal{S} = (D, M, H, E)$, Supernet $\mathcal{N}$ with maximum structural hyperparameters $(d_{\max}, m_{\max}, h_{\max}, e_{\max})$, Training dataset $\mathcal{D}$, Number of epochs T
Output: Trained supernet $\mathcal{N}$

```
Initialize N with random weights.
for epoch t = 1 to T do
    Randomly sample (d, m, h, e) from the predefined search space S, where:
        m = (m_1, m_2, ..., m_d) specifies the MLP ratio for each layer, and
        h = (h_1, h_2, ..., h_d) specifies the number of attention heads for each layer.
    Construct subnet N_s by selecting the first e embedding dimensions and the first d layers.
    for each layer l = 1 to d do
        Retain the first h_l attention heads and set MLP hidden dimensions to m_l × e.
    end for
    Extract weights W_s corresponding to N_s from the supernet N.
    for each mini-batch B ∈ D do
        Perform forward pass on N_s using W_s.
        compute loss L, and update W_s in N via gradient descent.
    end for
end for
return Trained supernet N.
```

C. Evolutionary Search

In the neural network architecture search process, based on (3), we aim to find the architecture that achieves the best performance on the validation set under a given parameter constraint

$$\begin{aligned} a^* &= \arg\max_{a\in\mathcal{A}} \text{acC}_{\text{val}}(\mathcal{N}(a, W_{\mathcal{A}}(a))) \\ &\text{s.t. } p_{a^*} \le C \end{aligned} \tag{7}$$

where p_{a^*} denotes the parameter quantity of a^*, and C denotes the given parameter constraint.

We employ an evolutionary algorithm to search for the target Transformer architecture. The search algorithm is presented in Algorithm 2, and the details of the crossover operator and mutation operator in the evolutionary algorithm will be provided in this section.

1) Crossover Operation: During the crossover operation, two encodings with the same length are selected from the population as parents. A new offspring is generated by uniform crossover. Fig. 5 shows an example of the crossover operator.

2) Mutation Operation: The evolutionary algorithm applies mutation to the depth and embedding dimension of an individual with a predefined probability P_s and mutates each remaining element with probability P_m. If the mutated network depth d' is less than the original depth d, the MLP ratio and the number of attention heads in the encoding inherit the first d' elements from the original. If the mutated depth d' is larger

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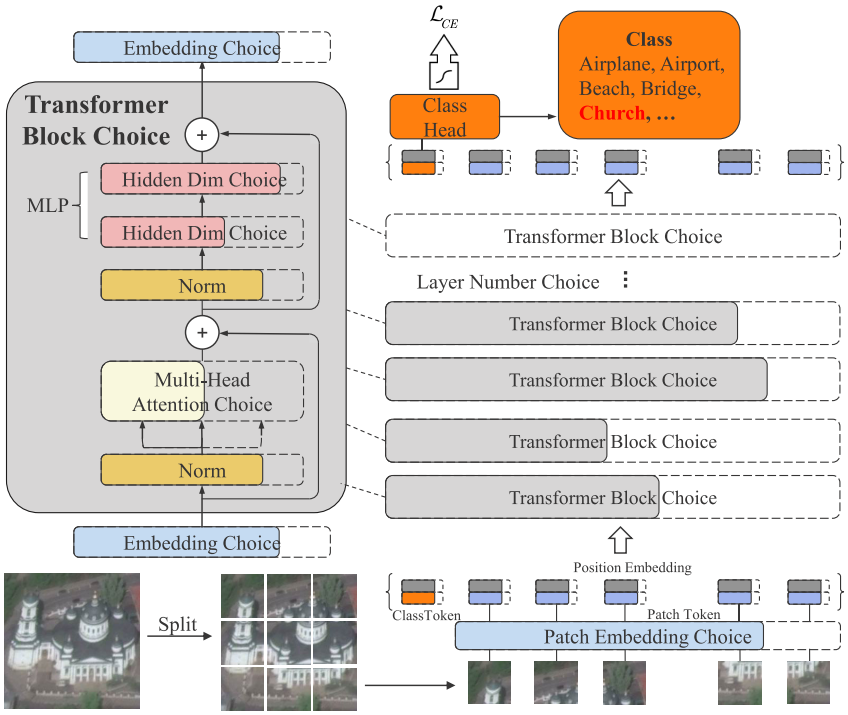


Fig. 3. Specific structure of the supernet. The dashed section in the figure represents the entire supernet. The solid section indicates a subnet, which is also a substructure of the supernet.

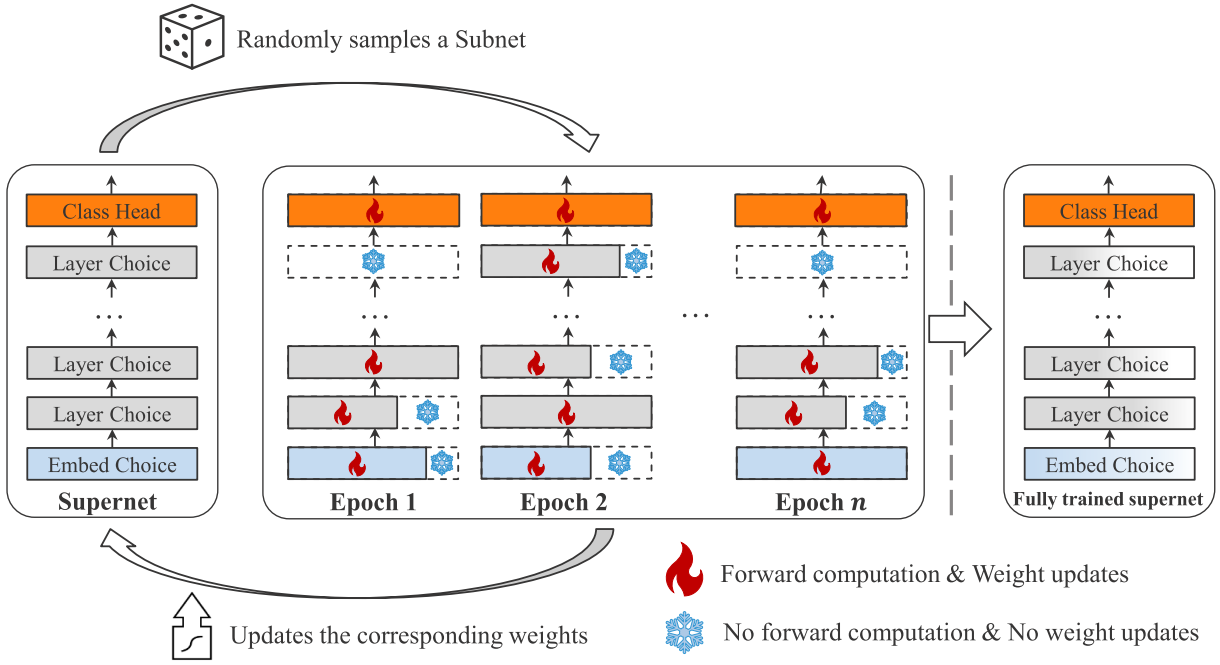


Fig. 4. Illustration of the supernet training strategy.

than the original depth d, the remaining $2 * (d' - d)$ elements are randomly filled from the search space. Fig. 5 provides an example of the mutation operator.

IV. Performance Evaluation

A. Implementation Details

1) Datasets: The supernet is pretrained on ImageNet-1k [35]. After pretraining, we finetune the supernet on the HSR image dataset to transfer it to scene classification tasks. The experiments on the HSR image dataset were conducted using fivefold cross-validation. The HSR image dataset includes the UCM dataset [36], the NWPU45 dataset [37], and the AID dataset [3]. Information about each dataset is provided in Table II.

2) Training Settings: We finetuned the pretrained supernet from AutoFormer on the HSR image datasets. We applied

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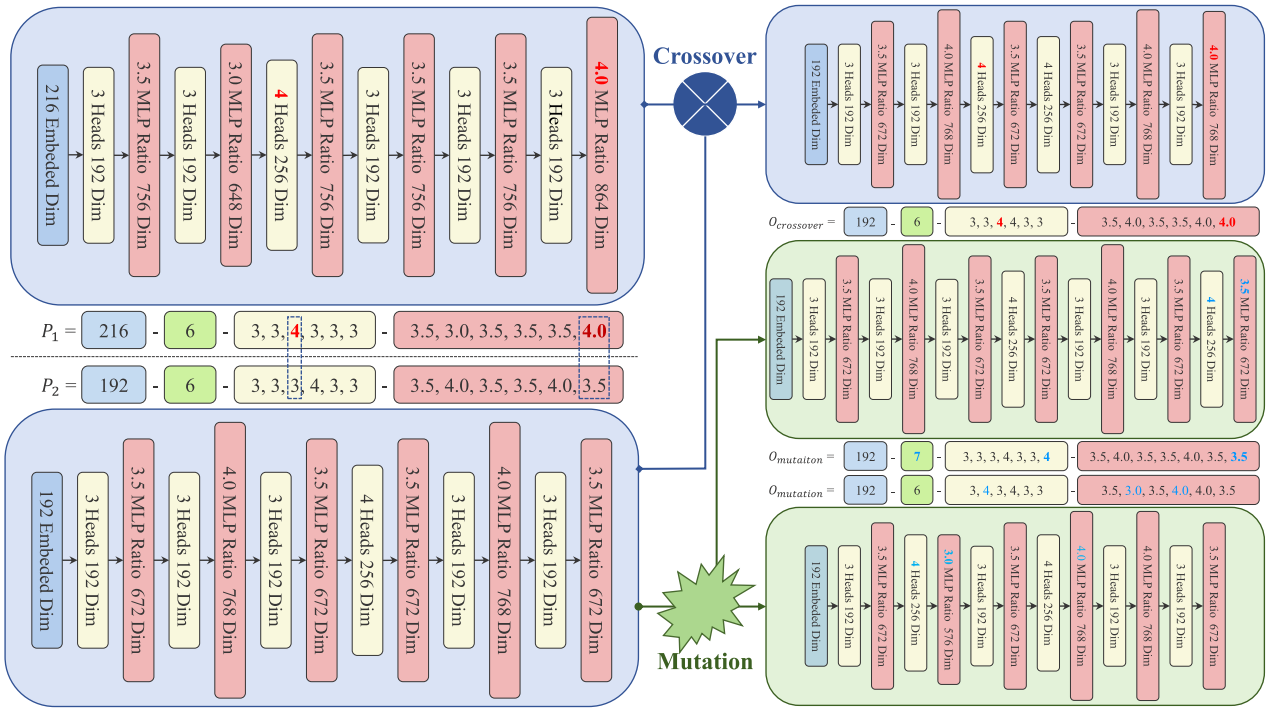


Fig. 5. Examples of crossover and mutation operators in evolutionary search.



Fig. 6. Examples of images from each category in the UCM dataset.

several data augmentation techniques, including RandAugment [38], Mixup [39], and Cutmix [40]. Table III provides the detailed configuration of these data augmentation methods and other training parameter settings. For all images in the dataset, we resized them to 224 × 224 pixels for model processing.

B. Experiment Results on UCM Dataset

The UC Merced (UCM) [36] dataset is a widely used standard benchmark dataset for remote sensing image classification tasks. The dataset is divided into 21 categories, including agricultural, beach, forest, golf course, harbor, river, etc. Each category contains 100 images. The resolution of each image is 256 × 256 pixels, with a spatial resolution of 0.3 meters. Fig. 6 shows some examples from the UCM dataset.

We use 80% of the dataset as the training set and 20% as the test set. Fig. 7 shows two neural network architectures that we obtained through a search on the

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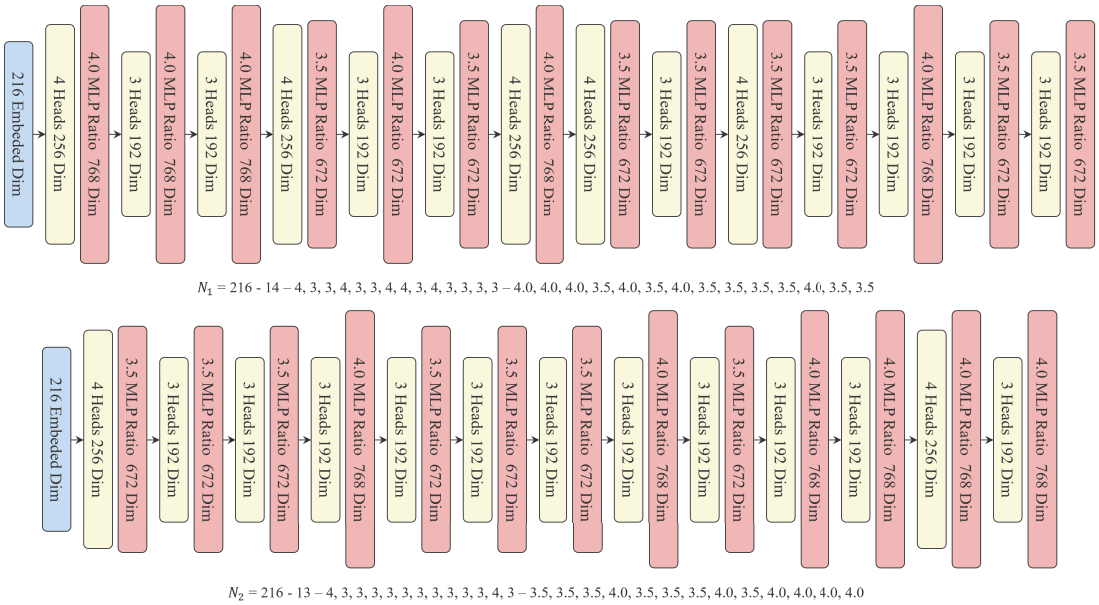


Fig. 7. Specific architectures of the two search results on the UCM dataset.

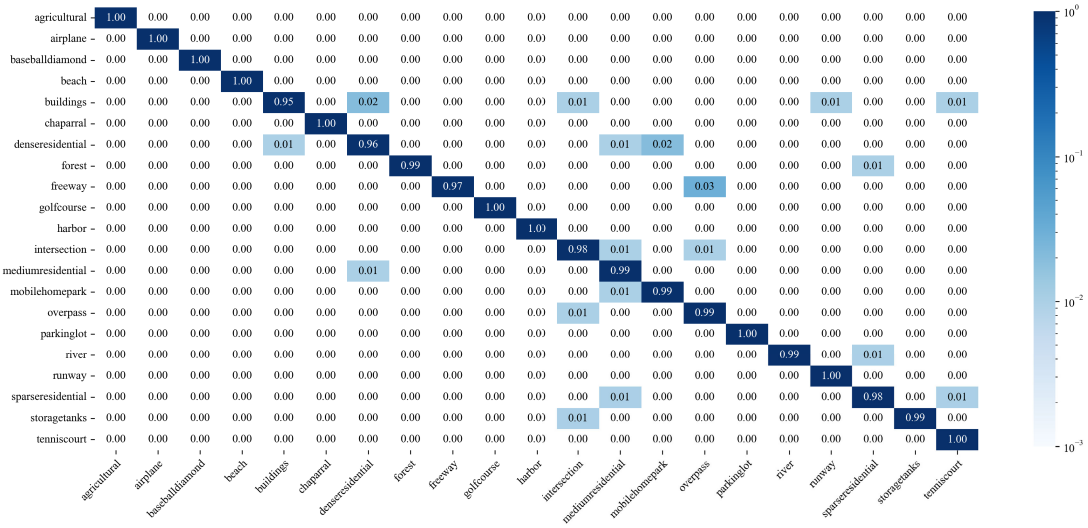


Fig. 8. Normalized confusion matrix of SceneFormer on the UCM dataset.

TABLE II

RELEVANT INFORMATION ABOUT THE DATASETS. AMONG THEM, IMAGENET-1K IS A NATURAL IMAGE DATASET, WHILE THE OTHER DATASETS ARE HSR IMAGE DATASETS

Dataset	Size	Classes	Images/Class	Image Size	Spatial Resolution
Imagenet-1k[35]	1.2M	1,000	∼1,200	256x256 (varies)	-
UCM[36]	2,100	21	100	256x256	∼0.3 m
NWPU45[37]	31,500	45	700	256x256	∼0.2 - 30 m
AID[3]	10,000	30	220-420	600x600	∼0.5 - 8 m

UCM dataset. Fig. 8 illustrates the confusion matrix of SceneFormer on the UCM dataset. Additionally, to demonstrate the effectiveness of SceneFormer, we compare it with several manually designed neural network architectures, including AlexNet [41], VGG16 [42], ResNet34 [43], and GoogleNet [44]. We also compare it with SceneNet [31], a CNN architecture search method for scene classification tasks. Table IV lists the experimental results on the UCM dataset. In Table IV, "Yes" under extra training data indicates that the model is pretrained on the natural image dataset ImageNet-1k.

The experimental results show that SceneFormer achieves competitive performance with lower computational costs. Specifically, compared to the CNN-based architecture search method SceneNet [31], we achieve similar classification performance on the UCM dataset with only 0.16× of its FLOPs. However, it is worth noting that the parameter quantity of SceneFormer remains substantially higher (6× that of RSNet [30], 7× that of SceneNet [31]). This is primarily due to the architectural characteristics of the Transformer. It relies on a large number of learnable parameters in its

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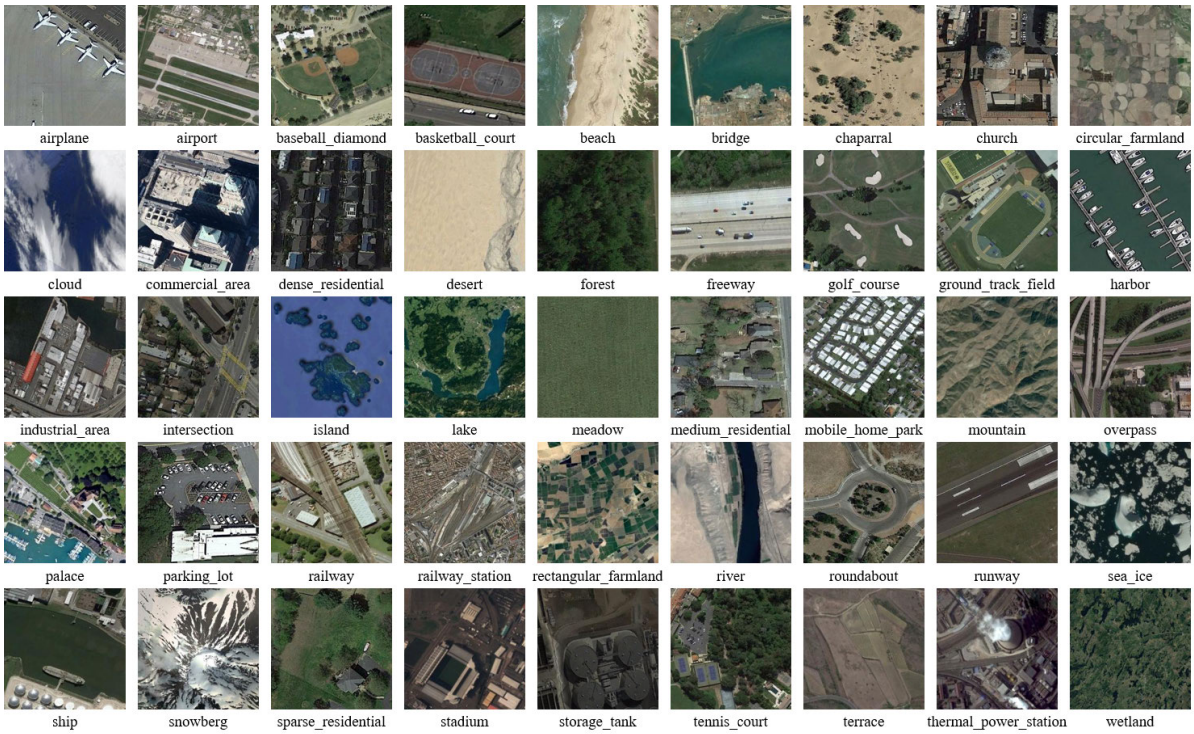


Fig. 9. Examples of images from each category in the NWPU45 dataset.

Algorithm 2 Evolutionary Search Algorithm

Input: population size N, parameter constraints $[c, C]$, the maximum generation number G, supernet training weights $W_{\mathcal{A}}$, validation dataset D_{val}
Output: the optimal architecture that satisfies the parameter quantity constraint in the search space
Initialize population P_0 with N random individuals satisfying $c \leq$ parameters $\leq C$
$Topk \leftarrow \emptyset$, $t \leftarrow 0$
while $t < G$ **do**
  Test the accuracy of P_t with $W_{\mathcal{A}}$ on the D_{val}.
  $Topk \leftarrow$ The top k individuals with the highest top-1 accuracy on D_{val} in $P_t \cup Topk$
  $P_{crossover} \leftarrow N/2$ individuals generated by the crossover operator that satisfy the parameter constraints.
  $P_{mutation} \leftarrow N/2$ individuals generated by the mutation operator that satisfy the parameter constraints.
  $t \leftarrow t + 1$.
  $P_t \leftarrow P_{crossover} \cup P_{mutation}$.
end while
return the individual in $Topk$ that satisfies the parameter constraint and achieves the highest accuracy.

attention and feedforward layers. Despite its higher parameter quantity, SceneFormer has lower computational cost and better accuracy. This makes it suitable for computationally resource-constrained scenarios.

TABLE III
DETAIL ABOUT THE TRAINING SETTINGS. THE TWO PARTS AROUND THE FORWARD SLASH (/) REPRESENT THE PARAMETER SETTINGS FOR TRAINING THE SUPERNET ON IMAGENET-1K AND FINETUNING ON THE HSR IMAGE DATASET, RESPECTIVELY. THE LAST ROW OF THE TABLE CORRESPONDS TO THE HYPERPARAMETERS FOR THE DATA AUGMENTATION METHODS USED. IN THE "RAND AUGMENT" COLUMN, THE SLASH SEPARATES THE NUMBER OF AUGMENTATION OPERATIONS AND THEIR PROBABILITY OF APPLICATION

Epochs	Batch size	Optimizer	Learning rate
500/ 300	1024/ 512	AdamW	1e-3/ 5e-4
Warmup epochs	Label smoothing	Learning rate decay	Weight decay
20	0.1	cosine	0.05
Rand Augment	Mixup prob.	Cutmix prob.	Erasing prob.
9/0.5	0.8	1.0	0.25

C. Experiment Results on NWPU45 Dataset

Compared with the UCM [36] dataset, the NWPU45 [37] dataset is a larger public dataset for remote sensing image classification. It contains 315 000 remote sensing images from different geographical regions. These images showcase a variety of scenes and environmental conditions. The dataset covers 45 scene categories, including airport, beach, bridge, church, golf course, lake, runway, etc. Each category contains 700 images, and each image measures 256 × 256 pixels. The spatial resolution of each image ranges from 20 cm to more than 30 m per pixel. The features extracted from images of different spatial resolutions can be inconsistent.

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TABLE IV
COMPARISON OF MODELS ON UCM DATASET. †: REPORTED BY [31], ‡: REPORTED BY [17]

Model	Params(M)	Flops(G)	OA(%)	kappa	design type	extra traing data
AlexNet†[41]	61.11	0.72	90.76 ± 0.72	0.8963 ± 0.0076	Manual	no
VGG16†[42]	138.38	15.45	96.76 ± 1.50	0.9669 ± 0.0177	Manual	no
ResNet34†[43]	21.26	4.15	95.48 ± 0.67	0.9524 ± 0.0071	Manual	no
GoogleNet†[44]	6.65	1.51	94.19 ± 0.65	0.9419 ± 0.0148	Manual	no
ResNet50‡[43]	25.6	-	96.98 ± 0.49	-	Manual	no
ViT-base‡[9]	86.4	-	99.01 ± 0.21	-	Manual	yes
PVT-Small‡[45]	24.0	-	98.81 ± 0.36	-	Manual	yes
Poolformer-Small‡[46]	30.4	-	99.02 ± 0.28	-	Manual	yes
HFFT-PD-PVT[17]	25.3	-	96.04 ± 0.28	-	Manual	yes
RSNet[30]	1.22	1.19	96.78 ± 0.60	0.9670 ± 0.0045	Auto	no
SceneNet[31]	1.08	10.03	99.10 ± 0.31	0.9905 ± 0.0033	Auto	no
E2SCNet[32]	3.99	0.48	98.76 ± 0.28	0.9871 ± 0.0028	Auto	no
SceneFormer(ours)	7.60 ± 0.27	1.69 ± 0.06	99.00 ± 0.28	0.9890 ± 0.0030	Auto	yes

TABLE V
COMPARISON OF MODELS ON NWPU45 DATASET. †: REPORTED BY [31], ‡: REPORTED BY [17]

Model	Params(M)	Flops(G)	OA(%)	kappa	design type	extra traing data
Results on Split-1 (80% training set, 20% testing set)						
AlexNet†[41]	61.11	0.72	87.91 ± 0.34	0.8140 ± 0.0034	Manual	no
VGG16†[42]	138.38	15.45	93.87 ± 0.44	0.9361 ± 0.0045	Manual	no
ResNet34†[43]	21.26	4.15	92.34 ± 0.63	0.9216 ± 0.0109	Manual	no
GoogleNet†[44]	6.65	1.51	92.58 ± 0.40	0.9225 ± 0.0041	Manual	no
RSNet[30]	1.40	1.28	92.64 ± 0.42	0.9247 ± 0.0041	Auto	no
SceneNet[31]	9.27	9.47	95.22 ± 0.22	0.9511 ± 0.0021	Auto	no
E2SCNet[32]	3.88	0.60	95.23 ± 0.16	0.9515 ± 0.0015	Auto	no
SceneFormer(ours)	7.55 ± 0.29	1.70 ± 0.07	97.51 ± 0.15	0.9743 ± 0.0017	Auto	yes
Results on Split-2 (10% training set, 90% testing set)						
ResNet50‡[43]	25.6	-	88.48 ± 0.21	-	Manual	no
ViT-base‡[9]	86.4	-	91.59 ± 0.19	-	Manual	yes
PVT-Small‡[45]	24.0	-	92.10 ± 0.35	-	Manual	yes
Poolformer-Small‡[46]	30.4	-	91.38 ± 0.22	-	Manual	yes
HFFT-PD-PVT[17]	25.3	-	92.79 ± 0.35	-	Manual	yes
SceneFormer(ours)	6.52 ± 0.24	1.51 ± 0.06	92.51 ± 0.18	0.9229 ± 0.20	Auto	yes

This inconsistency makes it challenging for models to adapt to varying input scales. As a result, it poses difficulties for scene classification tasks. Fig. 9 shows some category examples from the NWPU45 dataset.

We conducted experiments on the NWPU45 dataset using two different data splits: one with 80% of the images as the training set and the remaining 20% as the test set, and the other with 10% of the images as the training set and the remaining 90% as the test set. Compared with the UCM dataset, the NWPU dataset contains more scene categories and richer images. Experiments on this dataset further validate the Transformer's powerful capability to process global semantic information and demonstrate SceneFormer's adaptability to remote sensing scene classification tasks. The results show that compared to the CNN-based architecture search methods RSNet [30], SceneNet [31], and E2SCNet [32], SceneFormer achieves improvements of 4.87%, 2.29%, and 2.28% in overall classification accuracy on the NWPU45 dataset.

Fig. 10 presents the detailed structures of two search results on the NWPU45 dataset. Fig. 11 shows the normalized confusion matrix of SceneFormer on the NWPU45 dataset. Table V lists the experimental results on the NWPU45 dataset.

D. Experiment Results on AID Dataset

AID [3] dataset is a large-scale aerial image dataset. It is built from sample images collected from Google Earth imagery. The dataset contains 10 000 high-resolution aerial images, covering 30 aerial scene types, including airport, center, commercial, forest, playground, river, school, etc. Unlike the UCM dataset and the NWPU45 dataset, the number of images of each category in the AID dataset is uneven, ranging from 220 to 420. Additionally, the images in the AID dataset have a resolution of 600 × 600 pixels, containing more details and information. The spatial resolution of each image ranges from 50 cm to 8 m per pixel. These features present many challenges for scene classification tasks, such as class imbalance and the complexity of feature extraction from

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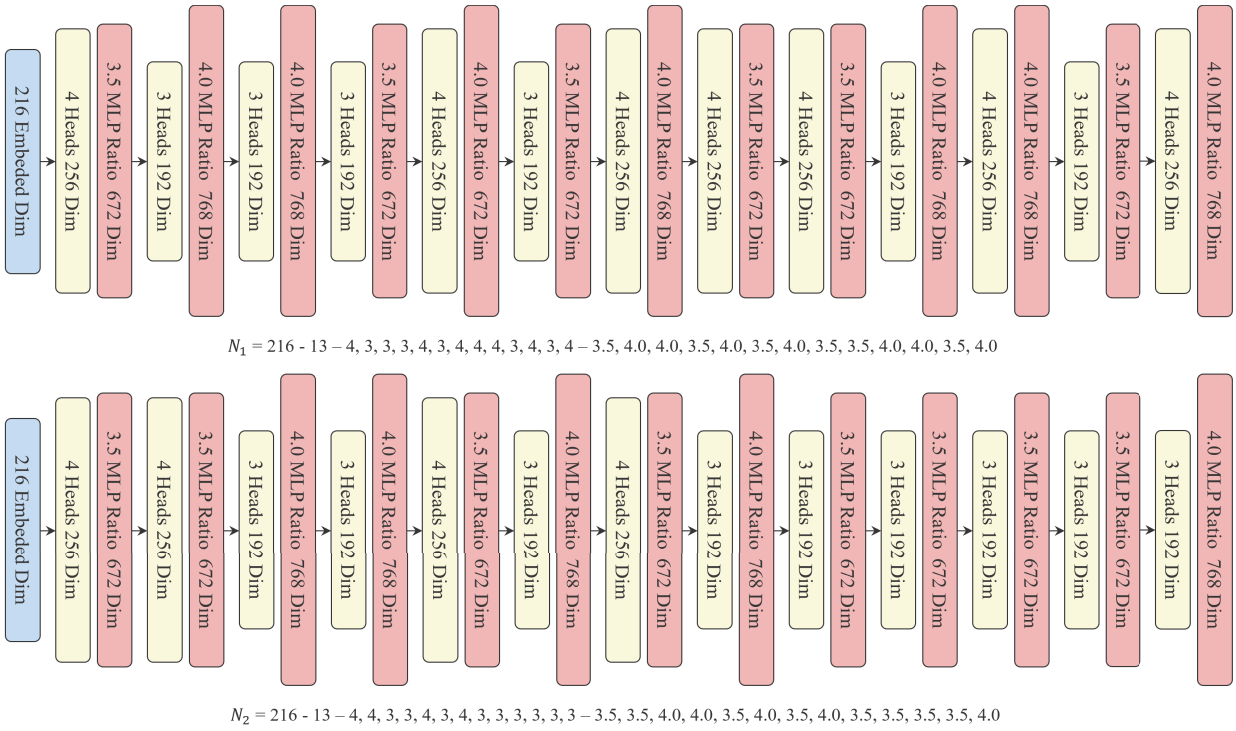


Fig. 10. Specific architectures of the two search results on the NWPU45 dataset.

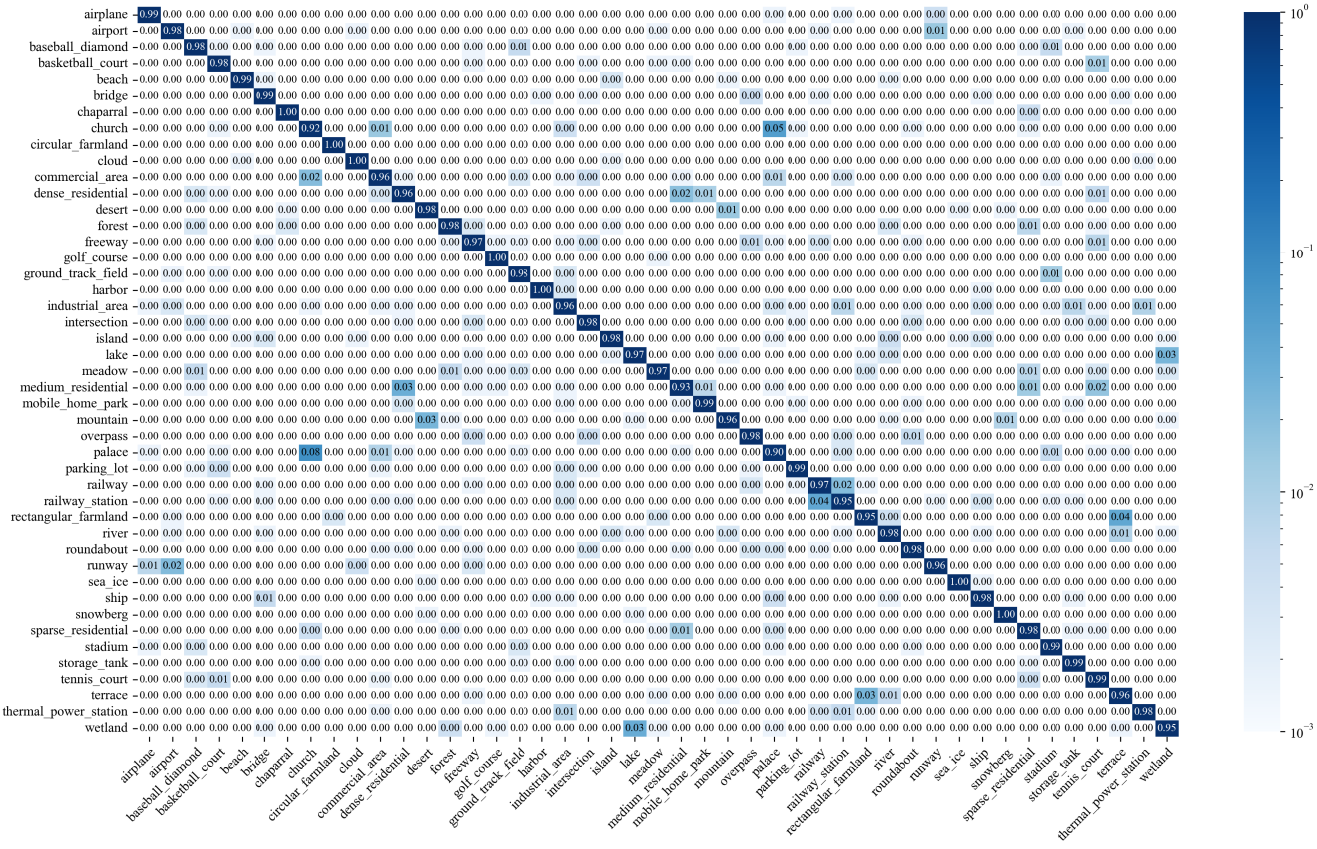


Fig. 11. Normalized confusion matrix of SceneFormer on the NWPU45 dataset.

high-resolution images. This requires researchers to explore and optimize their model designs further. Fig. 12 shows some examples from the AID dataset.

The AID dataset has the highest image resolution among the three datasets. SceneFormer demonstrates results significantly surpassing CNN-based methods on this dataset.

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Fig. 12. Examples of images from each category in the AID dataset.

TABLE VI
COMPARISON OF MODELS ON AID DATASET. †: REPORTED BY [31], ‡: REPORTED BY [17]

Model	Params(M)	Flops(G)	OA(%)	kappa	design type	extra traing data
AlexNet†[41]	61.11	5.04	81.90 ± 0.35	0.8157 ± 0.0047	Manual	no
VGG16†[42]	138.38	110.21	87.31 ± 0.96	0.8703 ± 0.0092	Manual	no
ResNet34†[43]	21.26	26.75	88.00 ± 0.67	0.8776 ± 0.0066	Manual	no
GoogleNet†[44]	6.65	10.73	83.81 ± 0.68	0.8357 ± 0.0053	Manual	no
ResNet50‡[43]	25.6	-	93.64 ± 0.25	-	Manual	no
ViT-base‡[9]	86.4	-	94.90 ± 0.29	-	Manual	yes
PVT-Small‡[45]	24.0	-	95.59 ± 0.16	-	Manual	yes
Poolformer-Small‡[46]	30.4	-	94.32 ± 0.26	-	Manual	yes
HFFT-PD-PVT[17]	25.3	-	96.04 ± 0.28	-	Manual	yes
RSNet[30]	1.55	8.70	87.55 ± 0.60	0.8705 ± 0.0059	Auto	no
SceneNet[31]	1.06	54.08	89.58 ± 0.29	0.8927 ± 0.0032	Auto	no
E2SCNet[32]	3.90	4.12	91.54 ± 0.31	0.9130 ± 0.0032	Auto	no
SceneFormer(ours)	7.48 ± 0.23	1.71 ± 0.04	96.14 ± 0.16	0.9593 ± 0.0018	Auto	yes

Specifically, SceneFormer improves overall classification accuracy by 6.56% on the AID dataset while using only 0.03× the FLOPs of SceneNet [31]. We attribute this success to two main factors. First, SceneFormer leverages the powerful representational capacity of the Transformer architecture by learning generic image features from natural image datasets. Second, compared to CNNs, Transformers are more effective at extracting global semantic information from standardized low-resolution images. By preprocessing high-resolution images into lower dimensions (224 × 224), SceneFormer efficiently reduces computational costs.

Fig. 13 presents the detailed structures of two search results on the AID dataset. Fig. 14 shows the normalized confusion matrix of SceneFormer on the AID dataset. Table VI lists the experimental results on the AID dataset, including comparisons with other Transformer model-based methods [9], [17], [45], [46].

E. Comparison of Search Cost With Existing NAS Methods for Remote Sensing Scene Classification

To evaluate the efficiency of our method, we compare its search cost with existing NAS methods designed for remote sensing scene classification. The search cost is measured in terms of GPU days. All our experiments are conducted on 4 NVIDIA RTX 2080 Ti GPUs. Table VII presents the comparison results. Our method significantly reduces the search cost while maintaining competitive or superior classification performance. Compared to other NAS methods for remote sensing scene classification, SceneFormer enables the subnets to directly inherit the supernet's weights after the search, thereby eliminating the need for retraining.

F. Parameter Analysis of the Mutation Probabilities P_s and P_m in the Evolutionary Algorithm

The mutation probabilities P_s and P_m are key parameters in the evolutionary algorithm. We conducted further analysis to assess the sensitivity of SceneFormer's performance to different hyperparameter settings. Specifically, we varied P_s and P_m from 0.1 to 0.5 in increments of 0.1, running the evolutionary algorithm multiple times and comparing the OA of the search results. The experimental results are presented in Table VIII, showing that the evolutionary algorithm achieves optimal performance when P_s is set to 0.4 and P_m to 0.2.

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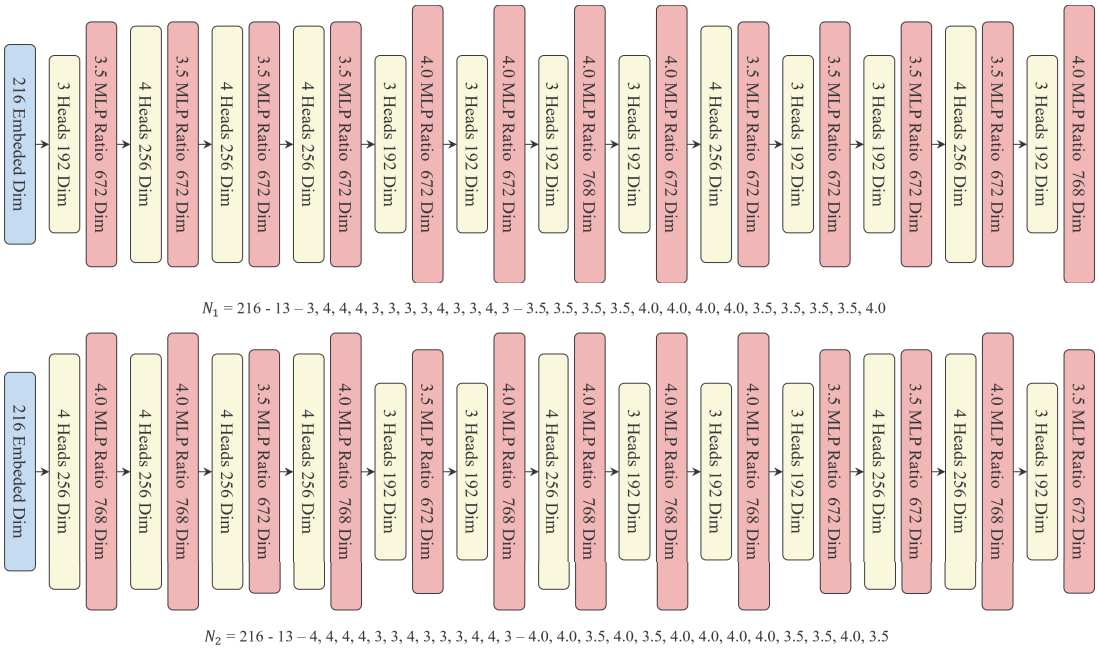


Fig. 13. Specific architectures of the two search results on the AID dataset.

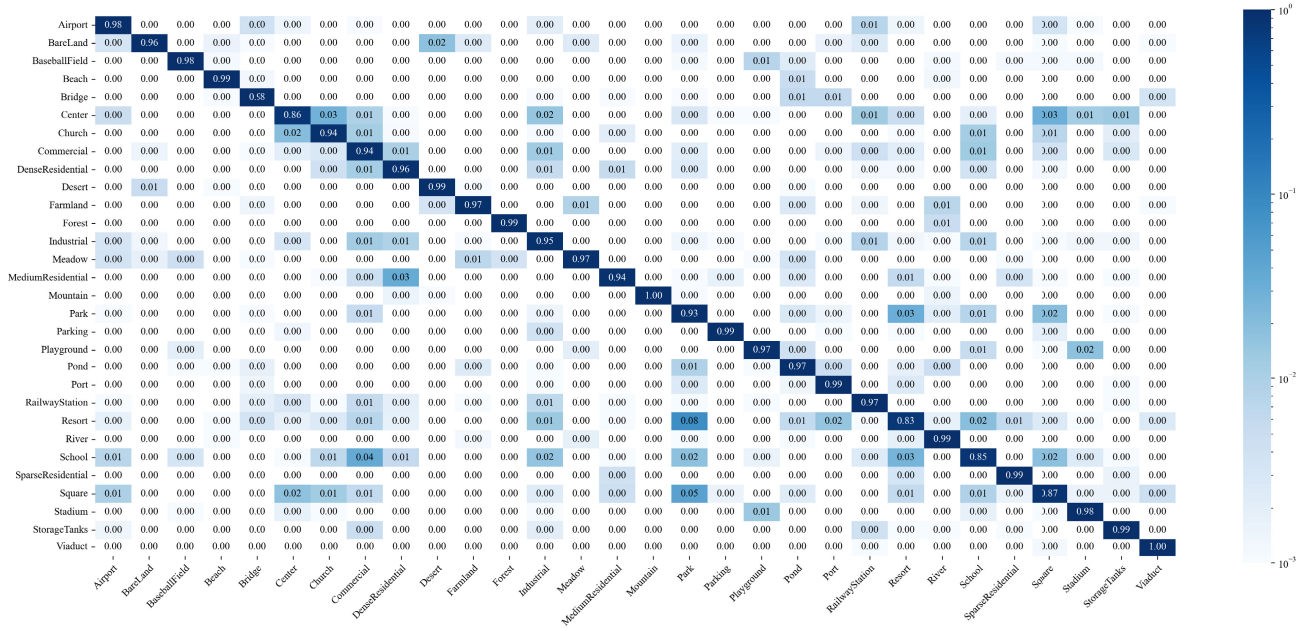


Fig. 14. Normalized confusion matrix of SceneFormer on the AID dataset.

TABLE VII

COMPARISON OF SEARCH COSTS. THE EXPERIMENTS USE 80%/80%/20% OF THE IMAGES FROM UCM/NWPU45/AID AS THE TRAINING SET, RESPECTIVELY. "REQUIRING RETRAINING" MEANS THAT THE SEARCH RESULTS NEED TO BE RETRAINED AFTER THE SEARCH IS COMPLETED. THE SEARCH COST OF OUR METHOD IS SPECIFICALLY DIVIDED INTO SUPERNET FINETUNING COST AND EVOLUTIONARY SEARCH COST, WHICH ARE PRESENTED ON EITHER SIDE OF THE "+" SYMBOL

Method	Search cost on UCM (GPU hours / GPU)	Search cost on NWPU45 (GPU hours / GPU)	Search cost on AID (GPU hours / GPU)	Requiring retraining	Search Method
RSNet[30]	24 / NVIDIA RTX 2080	48 / NVIDIA RTX 2080	72 / NVIDIA RTX 2080	yes	Gradient
SceneNet[31]	72 / NVIDIA RTX 2080	240 / NVIDIA RTX 2080	264 / NVIDIA RTX 2080	yes	Evolution
E2SCNet[32]	21 / NVIDIA RTX 2080	132 / NVIDIA RTX 2080	120 / NVIDIA RTX 2080	yes	Evolution
SceneFormer(ours)	2+4 / NVIDIA RTX 2080 Ti	10+10 / NVIDIA RTX 2080 Ti	2+12 / NVIDIA RTX 2080 Ti	no	Evolution

G. Ablation Study on Pretraining

To assess the impact of the pretraining process on our method, we conducted an ablation study by comparing the performance of the model with and without pretraining on the NWPU45 dataset. As presented in Table IX, pretraining improves the OA of SceneFormer by 1.71%.

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TABLE VIII
IMPACT OF MUTATION PROBABILITIES P_s AND P_m ON SCENEFORMER'S PERFORMANCE

P_s (P_m = 0.2)	0.1	0.2	0.3	**0.4**	0.5
OA(%)	98.47	98.38	98.57	**98.76**	98.57
P_m (P_s = 0.4)	0.1	**0.2**	0.3	0.4	0.5
OA(%)	98.57	**98.76**	98.67	98.48	98.38

TABLE IX
PERFORMANCE COMPARISON WITH AND WITHOUT PRETRAINING

	Params(M)	Flops(G)	OA(%)	kappa
w/o pre-train	7.61 ± 0.20	1.71 ± 0.04	95.80 ± 0.19	0.9569 ± 0.0019
pre-train	7.55 ± 0.29	1.70 ± 0.07	97.51 ± 0.15	0.9743 ± 0.0017

This demonstrates the effectiveness of leveraging large-scale datasets for feature initialization.

V. CONCLUSION

This article presents SceneFormer, a ViT specifically designed for remote sensing scene classification. We design a search space for finding the optimal Transformer architecture with a heterogeneous layer design. To improve search efficiency, we introduce a supernet-based framework, which allows for the effective evaluation of candidate architectures. The evolutionary algorithm guides the search for the optimal Transformer architecture under specific parameter constraints. The proposed framework provides a promising architecture design method for resource-constrained environments, where the tradeoffs between accuracy and efficiency are crucial. Experiments on UCM, NWPU45, and AID datasets demonstrate SceneFormer's superior accuracy, showcasing its effectiveness in automating Transformer design for remote sensing classification tasks.

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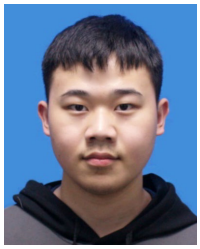
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